

Automated minus reference counts on the gold standard: protocol change

tranche mixed_gold_02 - frozen baseline versus candidate protocol on the same manifest and gold - misses and abstentions scored as 0 - square root axes - every asserted gold row

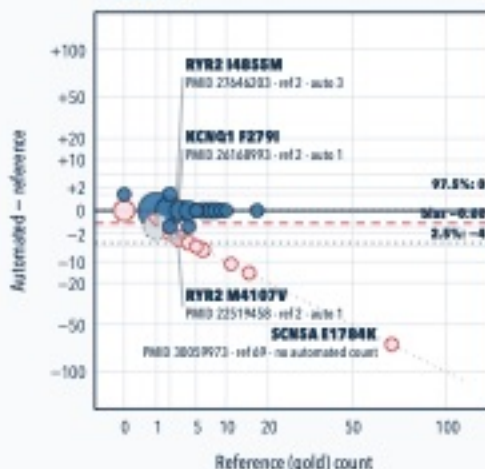
● automated count supplied ○ variant matched, no automated count (scored 0) ○ variant not found (scored 0)

Marker area - gold rows at one coordinate (wedges split it by status):

○ 1 ○ 8 ○ 28

Baseline arm - 20260903_protocol_mixed02_baseline - runtime 8235a53c - locked 2026-09-03 - 49 gene-paper attempts

Carriers



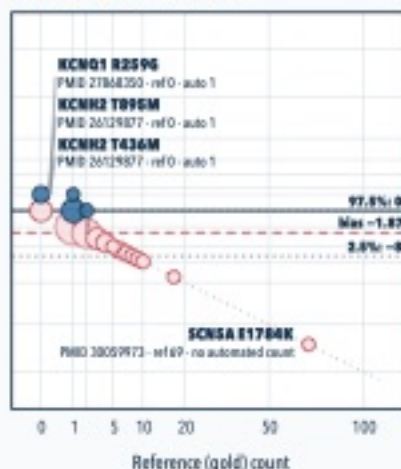
303 gold rows - matched 88.4% - supplied 76.6%

exact 80.2% (zero-baseline 5.6%) - within ± 1 94.4%

bias -0.60 - c2e MAE 0.620 - cond. MAE 0.026 (n 232)

empirical 95% limits -4 to 0 - RMSE 4.20

Affected individuals



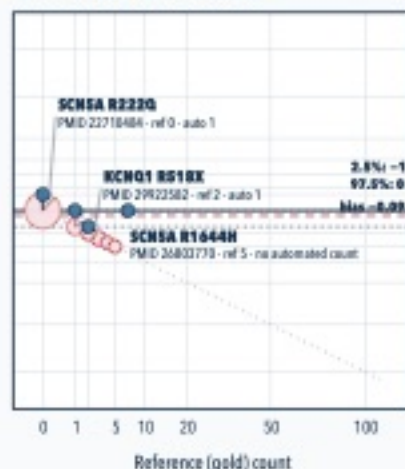
303 gold rows - matched 88.4% - supplied 11.6%

exact 16.2% (zero-baseline 6.9%) - within ± 1 65.3%

bias -1.87 - c2e MAE 1.921 - cond. MAE 0.200 (n 35)

empirical 95% limits -8 to 0 - RMSE 4.74

Unaffected individuals



303 gold rows - matched 88.4% - supplied 2.0%

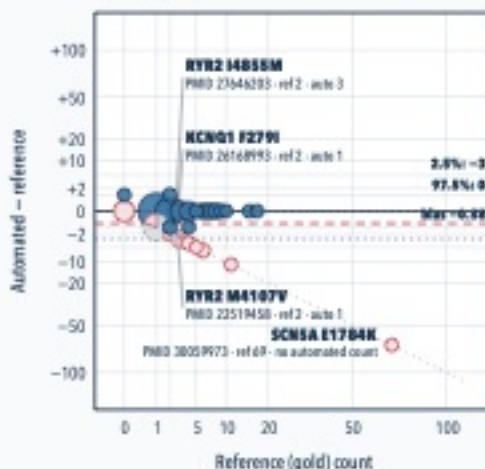
exact 95.0% (zero-baseline 93.7%) - within ± 1 97.7%

bias -0.09 - c2e MAE 0.099 - cond. MAE 0.333 (n 6)

empirical 95% limits -1 to 0 - RMSE 0.53

Candidate arm - 20260903_protocol_mixed02_candidate - runtime 0f1d6e65 - locked 2026-09-03 - 49 gene-paper attempts

Carriers



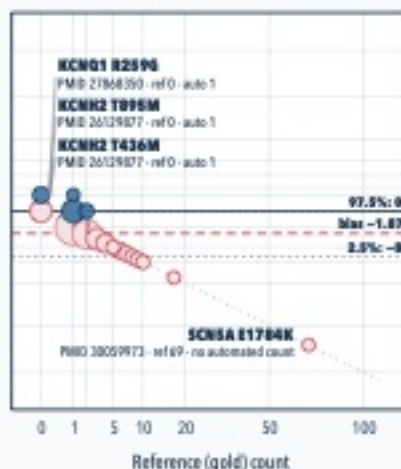
303 gold rows - matched 88.1% - supplied 74.9%

exact 78.5% (zero-baseline 5.6%) - within ± 1 94.4%

bias -0.58 - c2e MAE 0.597 - cond. MAE 0.026 (n 232)

empirical 95% limits -3 to 0 - RMSE 4.12

Affected individuals



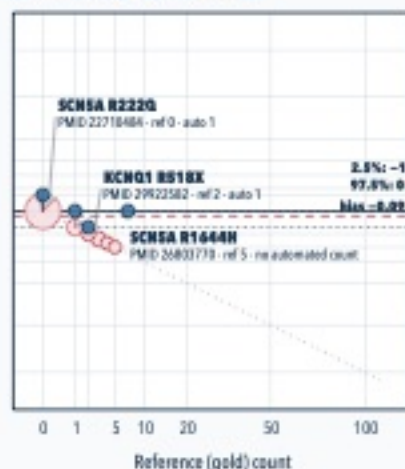
303 gold rows - matched 88.1% - supplied 11.2%

exact 15.8% (zero-baseline 6.9%) - within ± 1 65.3%

bias -1.87 - c2e MAE 1.921 - cond. MAE 0.206 (n 34)

empirical 95% limits -8 to 0 - RMSE 4.74

Unaffected individuals



303 gold rows - matched 88.1% - supplied 2.3%

exact 95.0% (zero-baseline 93.4%) - within ± 1 97.7%

bias -0.09 - c2e MAE 0.099 - cond. MAE 0.286 (n 7)

empirical 95% limits -1 to 0 - RMSE 0.53

Candidate - baseline on the same 909 gold rows (paired by gold row; deltas are candidate minus baseline)

Carriers

c2e MAE 0.620 $\rightarrow$ 0.597 (-0.023) - registered UB +0.037

supplied 232 $\rightarrow$ 227 - exact 243 $\rightarrow$ 238 - bias -0.60 $\rightarrow$ -0.58

rows better 1 - worse 6 - unchanged 296

coverage on matched 86.6% $\rightarrow$ 83.8% (-1.55 pp) - LB -9.54 pp

Affected individuals

c2e MAE 1.921 $\rightarrow$ 1.921 (± 0.000)

supplied 35 $\rightarrow$ 34 - exact 49 $\rightarrow$ 48 - bias -1.87 $\rightarrow$ -1.87

rows better 2 - worse 3 - unchanged 298

coverage on matched 13.1% $\rightarrow$ 12.7% (-0.33 pp)

Unaffected individuals

c2e MAE 0.099 $\rightarrow$ 0.099 (± 0.000)

supplied 6 $\rightarrow$ 7 - exact 288 $\rightarrow$ 288 - bias -0.09 $\rightarrow$ -0.09

rows better 0 - worse 0 - unchanged 303

coverage on matched 2.2% $\rightarrow$ 2.6% (+0.38 pp)

Identity (primary endpoint): recall 88.45% $\rightarrow$ 88.12% (-0.33 pp; LB -1.89 pp) - precision 83.23% $\rightarrow$ 82.92% (-0.31 pp; LB -1.65 pp) - registered decision: reject_or_revise_candidate - secondary counts